BIO-RAD

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Species Used *Lactobacillus acidophilus* ADH; gastrointestinal isolate from human faeces

Molecules Electroporated DNA: pGT633, covalently closed circular form, a native 9.8kb erythromycin resistant *Lactobacillus* plasmid.

Before the Pulse

Cell Growth Medium Lactobacilli MRS broth (Difco)

Growth Phase at Harvest O.D. (600) =log phase cells, 0.8

Pre-pulse Incubation 0°C for 1 min.

Wash Solution 3.5X SMEB (Luchansky *et.al.* 1988 BioRad Bulletin 1350:1-3)

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium 3.5X SMEB (1x = 272 mM sucrose, 1 mM MgCl2)

Cell Density 10 (9) cells / ml

Volume of Cells 800 µl

DNA Concentration 10 µg

DNA Resuspension Buffer TE (10 mM Tris, 1 mM EDTA)

Volume of DNA 5 µl

Cuvette Gap 0.4 cm

Voltage 2.5 kV

Field Strength 6.25 kV/cm

Capacitor 25 µF

Resistor Pulse Controller not used.

Time Constant 10 to 15 msec

After the Pulse

Outgrowth Medium Lactobacilli MRS broth (Difco) 10ml

Outgrowth Temperature 37 °C

Length of Incubation 3 hrs.

Selection Method or Assay Used Erythromycin, 25 µg/ml; the Em(R) gene of pGTG33 requires a min.expression time of 3hr.to recover transformants

Electroporation Efficiency Average 8.6 x 10 (1) transformants / µg DNA

Per Cent Survival 17%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
It is NOT RECOMMENDED to use high voltage with out the Pulse Controller. **Ref: (1) H. J. Connell, "Investigation of Methods for the Transformation of Gastointestinal Strains of Lactobacilli with Plasmid pGT633." Ph.D. Thesis, University of Otago, Dunedin, NEW ZEALAND (1990) **(2)** This work was carried out under the supervision of Dr. G. Tannock , Dept. of Microbiol., Univ. of Otago, P. O. Box 56, Dunedin, NEW ZEALAND; PH:+64-3-4797713; FAX: 64-3- 4741607. Questions regarding the availability of strains and the plasmid pGT633 should be directed to him.

Name of Submittor Dr. H. J. Connell

Institution Address Lund University,
Clinical Immunology, Dept. of Medical Microbiology,
Solvegatan 23,Sund S-22362
Sweden

Survey Number 062